1 (a) $\overline{A \cap B} = \bar{A} \cup \bar{B}$

let $x \in \overline{A \cap B}$

$\Rightarrow x \notin A \cap B$

$\Rightarrow x \notin A$ or $x \notin B$

$\Rightarrow x \in \bar{A}$ or $x \in \bar{B}$

$\Rightarrow x \in \bar{A} \cup \bar{B}$

$\Rightarrow \overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$ —— (1)

now let $x \in \bar{A} \cup \bar{B}$

$\Rightarrow x \in \bar{A}$ or $x \in \bar{B}$

$\Rightarrow x \notin A$ or $x \notin B$

$\Rightarrow x \notin A \cap B$

$\Rightarrow x \in \overline{A \cap B}$

$\Rightarrow \bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$ —— (2)

by (1) & (2), $\overline{A \cap B} = \bar{A} \cup \bar{B}$

(b) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

let $x \in \overline{A \cup B}$

$\Rightarrow x \notin A \cup B$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow$ ~~$x \notin \bar{A}$ and $x \notin \bar{B}$~~ $x \in \bar{A}$ and $x \in \bar{B}$

$\Rightarrow x \in \bar{A} \cap \bar{B}$

$\Rightarrow \overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$ —— (1)

now let $x \in \bar{A} \cup \bar{B}$

$\Rightarrow x \in \bar{A}$ and $x \in \bar{B}$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \notin A \cup B$

$\Rightarrow x \in \overline{A \cup B}$

$\Rightarrow \bar{A} \cap \bar{B} \subseteq \overline{A \cup B}$ —— (2)

by (1) & (2), $\overline{A \cup B} = \bar{A} \cap \bar{B}$ Proved

② In a crisp set, each element either belongs to the set or does not belong.

So, membership values are $\mu_A(x) = \begin{cases} 1 & , x \in A \\ 0 & , x \notin A \end{cases}$

This converts a crisp set into a fuzzy set with only 0 and 1 values

So equivalent fuzzy set is $\{(1,1),(2,0),(3,1),(5,0),(7,1)\}$.

③ Given $\mu(x) = \frac{x}{1-x} \; \forall x \in \mathbb{R}$.

For a function to qualify as a membership function, it must satisfy $0 \leq \mu(x) \leq 1 \;\; \forall x$.

Given $\mu(x)$ is undefined at $x = 1$.

So $\mu(x)$ does not qualify as a membership function.

(4) $X = \{10, 20, 30, 40, 50\}$

$$\mu_A(x) = \frac{x}{x+10}$$

$$\mu_B(x) = \frac{2x}{3x+10}$$

So, $\mu_A(10) = \frac{10}{20} = .5$

$\mu_A(20) = \frac{20}{30} = .66$

$\mu_A(30) = \frac{30}{40} = .75$

$\mu_A(40) = \frac{40}{50} = .80$

$\mu_A(50) = \frac{50}{60} = .83$

$\mu_B(10) = \frac{20}{40} = .5$

$\mu_B(20) = \frac{40}{70} = .57$

$\mu_B(30) = \frac{60}{100} = .60$

$\mu_B(40) = \frac{80}{130} = .61$

$\mu_B(50) = \frac{100}{160} = .62$

(a) So,

$A = \{(10, .5), (20, .66), (30, .75), (40, .80), (50, .83)\}$

$B = \{(10, .5), (20, .57), (30, .60), (40, .61), (50, .62)\}$

(b) and (c)

We know

$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

$\mu_{A^c}(x) = 1 - \mu_A(x)$

So,

$A \cap B = \{(10, .5), (20, .57), (30, .60), (40, .61), (50, .62)\}$

$A \cup B = \{(10, .5), (20, .66), (30, .75), (40, .80), (50, .83)\}$

also we know

$$|A| = \sum_{x \in X} \mu_A(x)$$

$|A \cap B| = .5 + .57 + .60 + .61 + .62 = 2.9$

$|A \cup B| = .5 + .66 + .75 + .80 + .83 = 3.54$

5. $X = \{a, b, c, d\}$

$A = \{(a, .2), (b, .7), (c, 1), (d, .9)\}$

$B = \{(a, .1), (b, .8), (d, .2)\}$

$A \cap B = \{(a, .1), (b, .7), (d, .2)\}$

$|A \cap B| = .1 + .7 + .2 = 1$

$A \cup B = \{(a, .2), (b, .8), (c, 1), (d, .9)\}$

$|A \cup B| = .2 + .8 + 1 + .9 = 2.9$

$A^c = \{(a, .8), (b, .3) (c, 0), (d, .1)\}$

$|A^c| = .8 + .3 + 0 + .1$

$= 1.2.$

(6) $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$

We have

$\overline{A \cup (B \cap C)} = \bar{A} \cap (\overline{B \cap C})$ by De morgan law

$= \bar{A} \cap (\bar{B} \cup \bar{C})$ " "

$= (\bar{B} \cup \bar{C}) \cap \bar{A}$ by commutative law for intersections

$= (\bar{C} \cup \bar{B}) \cap \bar{A}$ by commutative law for unions.

(7) $A_i = \{1, 2, 3, \dots i\}$ for $i = 1, 2, 3 \dots$ Then,

$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \{1, 2, 3, \dots i\} = \{1, 2, 3 \dots\} = Z^+$

and

$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} \{1, 2, 3 \dots i\} = \{1\}.$

(8) $\overline{A \cap B} = \bar{A} \cup \bar{B}$

take, $\overline{A \cap B} = \{x \mid x \notin A \cap B\}$

$= \{x \mid \sim(x \in (A \cap B))\}$

$= \{x \mid \sim(x \in A \wedge x \in B)\}$ by def^n of intersection

$= \{x \mid \sim(x \in A) \vee \sim(x \in B)\}$ by Demorgan law for logical equivalences

$= \{x \mid x \notin A \vee x \notin B\}$ by def^n of does not belong symbol

$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$ by def^n of complement

$= \{x \mid x \in \bar{A} \cup \bar{B}\}$ by def^n of union.

$= \bar{A} \cup \bar{B}.$ by meaning of set builder notation.